

第二章 行列式(1)

知识点 线性方程组行列式解法、 n 阶行列式的定义与性质

课后练习

1. 1) $\begin{vmatrix} 2 & 5 \\ 3 & 7 \end{vmatrix} = 14 - 15 = -1$

2) $\begin{vmatrix} a & a^2 \\ b & b^2 \end{vmatrix} = ab^2 - a^2b = ab(b-a)$

3) $\begin{vmatrix} 1 & 2 & 5 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{vmatrix} = 6$

4) $\begin{vmatrix} 0 & 0 & x \\ 0 & 2 & x \\ x & -1 & 3 \end{vmatrix} = (-1)^{z(3,2,1)} \cdot 2x^2 = -2x^2$

5) $\begin{vmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{vmatrix} = (-1)^{z(2,1,4,3)} = 1$

6) $\begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & 0 & 0 & 0 \\ a_{41} & a_{42} & 0 & 0 & 0 \\ a_{51} & a_{52} & 0 & 0 & 0 \end{vmatrix} = 0$

$\begin{vmatrix} C & B \\ A & O \end{vmatrix}$

2. 排列 $\overset{0}{5} \overset{1}{3} \overset{2}{7} \overset{3}{1} \overset{4}{2} \overset{5}{4} \overset{6}{6}$ 的逆序数为 0, 排列 $\overset{n}{1}, \overset{n-1}{3}, \dots, \overset{2}{(2n-1)}, \overset{1}{2}, \overset{0}{4}, \dots, \overset{0}{2n}$ 的逆序数为 $\frac{n(n-1)}{2}$, 六阶行列式中 $a_{13}a_{25}a_{36}a_{41}a_{54}a_{62}$ 的符号为 $(-1)^{z(3,5,6,1,4,2)} = -1$.

3. 已知 A, B 均为 4 阶矩阵, 且 $|A|=2, |B|=3$, 则 $|AB^T| = |A||B^T| = 6$

$|A|B| = |2B| = 2^4|B| = 48$

4. 设多项式 $f(x) = \begin{vmatrix} 2x & x & 2 & 3 \\ 1 & x & 2 & 4 \\ 3 & 2 & 3x & -1 \\ 2 & 5 & 3 & x \end{vmatrix}$, 则 $f(x)$ 中 x^4 的系数为 6, x^3 的系数

为 -3, 常数项为 $f(0) = \begin{vmatrix} 0 & 0 & 2 & 3 \\ 1 & 0 & 2 & 4 \\ 3 & 2 & 0 & -1 \\ 2 & 5 & 3 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 2 & 4 \\ 3 & 2 & 0 & -1 \\ 0 & 0 & 2 & 3 \\ 2 & 5 & 3 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 2 & 4 \\ 0 & 2 & -6 & -13 \\ 0 & 0 & 2 & 3 \\ 0 & 5 & -1 & -8 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 2 & 4 \\ 0 & 2 & -6 & -13 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 14 & \frac{49}{2} \end{vmatrix}$

5. 利用行列式的性质计算下列行列式:

$= \begin{vmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \end{vmatrix} \cdot \begin{vmatrix} 2 & 3 \\ 14 & \frac{49}{2} \end{vmatrix} = 14$

1) $\begin{vmatrix} a & b & c & d \\ b & c & a & 1 \\ c & a & b & 1 \\ \frac{b+c}{2} & \frac{c+a}{2} & \frac{a+b}{2} & 1 \end{vmatrix} = 0$; $\begin{vmatrix} 0 & a_{12} & a_{13} & a_{14} & a_{15} \\ -a_{12} & 0 & a_{23} & a_{24} & a_{25} \\ -a_{13} & -a_{23} & 0 & a_{34} & a_{35} \\ -a_{14} & -a_{24} & -a_{34} & 0 & a_{45} \\ -a_{15} & -a_{25} & -a_{35} & -a_{45} & 0 \end{vmatrix} = 0$;

2) 设 $\begin{vmatrix} a_1 & c_1 & d_1 \\ a_2 & c_2 & d_2 \\ a_3 & c_3 & d_3 \end{vmatrix} = 4$, $\begin{vmatrix} b_1 & c_1 & d_1 \\ b_2 & c_2 & d_2 \\ b_3 & c_3 & d_3 \end{vmatrix} = 1$, 则 $\begin{vmatrix} a_1+b_1 & 2d_1 & 3c_1 \\ a_2+b_2 & 2d_2 & 3c_2 \\ a_3+b_3 & 2d_3 & 3c_3 \end{vmatrix} = 6 \begin{vmatrix} a_1+b_1 & d_1 & c_1 \\ a_2+b_2 & d_2 & c_2 \\ a_3+b_3 & d_3 & c_3 \end{vmatrix} = 6 \cdot (-4 - 1) = -30$

$$3) \begin{vmatrix} 0 & \cdots & 0 & a_1 \\ 0 & \cdots & a_2 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ a_n & \cdots & 0 & 0 \end{vmatrix} = \frac{n(n-1)}{2} a_1 a_2 \cdots a_n; \quad \begin{vmatrix} 1 & 2 & 0 & 0 \\ 3 & 4 & 0 & 0 \\ 2 & 5 & 2 & -1 \\ 6 & -3 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} \cdot \begin{vmatrix} 2 & -1 \\ 1 & 0 \end{vmatrix} = -2 \cdot 1 = -2$$

$$4) \begin{vmatrix} 103 & 100 & 204 \\ 199 & 200 & 395 \\ 301 & 300 & 600 \end{vmatrix} = 100 \begin{vmatrix} 3 & 1 & 4 \\ -1 & 2 & -5 \\ 1 & 3 & 0 \end{vmatrix} = 100 (0 + (-5) + (-12) - 8 - 0 - (-45)) = 2000$$

$$5) \begin{vmatrix} -ab & ac & ae \\ bd & -cd & de \\ bf & cf & -ef \end{vmatrix} = adfbce \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} = abcdef \begin{vmatrix} -1 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \end{vmatrix} = 4 abcdef$$

$$6) \begin{vmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{vmatrix} = \begin{vmatrix} 10 & 2 & 3 & 4 \\ 10 & 3 & 4 & 1 \\ 10 & 4 & 1 & 2 \\ 10 & 1 & 2 & 3 \end{vmatrix} = 10 \begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & -3 \\ 0 & 2 & -2 & -2 \\ 0 & -1 & -1 & -1 \end{vmatrix} = 20 \begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & -3 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & -4 \end{vmatrix} = 160$$

$$7) \begin{vmatrix} 2 & 1 & 4 & 1 \\ 1 & 2 & 3 & 2 \\ 3 & -1 & 2 & 1 \\ 5 & 0 & 6 & 2 \end{vmatrix} \xrightarrow{r_1 \leftrightarrow r_2} \begin{vmatrix} 1 & 2 & 3 & 2 \\ 2 & 1 & 4 & 1 \\ 3 & -1 & 2 & 1 \\ 5 & 0 & 6 & 2 \end{vmatrix} \xrightarrow{r_2 - 2r_1, r_3 - 3r_1, r_4 - 5r_1} \begin{vmatrix} 1 & 2 & 3 & 2 \\ 0 & -3 & -2 & -3 \\ 0 & -7 & -7 & -5 \\ 0 & -10 & -9 & -8 \end{vmatrix} \xrightarrow{r_2 \leftrightarrow r_4} \begin{vmatrix} 1 & 2 & 3 & 2 \\ 0 & -10 & -9 & -8 \\ 0 & -7 & -7 & -5 \\ 0 & -3 & -2 & -3 \end{vmatrix} \xrightarrow{r_2 \leftrightarrow r_4} \begin{vmatrix} 1 & 2 & 3 & 2 \\ 0 & -3 & -2 & -3 \\ 0 & -7 & -7 & -5 \\ 0 & -10 & -9 & -8 \end{vmatrix} = -2 \begin{vmatrix} -3 & 1 & 0 \\ -7 & 0 & 1 \\ -10 & 1 & 1 \end{vmatrix} = -2(-10 - (-7) - (-3)) = 0$$

或者观察 $\rightarrow \begin{vmatrix} 2 & 1 & 4 & 1 \\ 1 & 2 & 3 & 2 \\ 5 & 0 & 6 & 2 \\ 5 & 0 & 6 & 2 \end{vmatrix} = 0$

6. 用性质证明下列等式

(1)

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ bc & ca & ab \end{vmatrix} = (a-b)(b-c)(c-a)$$

$$\begin{vmatrix} 1 & 0 & 0 \\ a & b-a & c-a \\ bc & c(a-b) & b(a-c) \end{vmatrix} = (a-b)(c-a) \begin{vmatrix} 1 & 0 & 0 \\ a & -1 & 1 \\ bc & c & -b \end{vmatrix}$$

$$\stackrel{c_2+c_1}{=} (a-b)(c-a) \begin{vmatrix} 1 & 0 & 0 \\ a & -1 & 0 \\ bc & c & c-b \end{vmatrix} = (a-b)(b-c)(c-a)$$

$$(2) \begin{vmatrix} b_1+c_1 & a_1+c_1 & b_1+a_1 \\ b_2+c_2 & a_2+c_2 & b_2+a_2 \\ b_3+c_3 & a_3+c_3 & b_3+a_3 \end{vmatrix} = 2 \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$= \begin{vmatrix} b_1 & a_1+c_1 & b_1+a_1 \\ b_2 & a_2+c_2 & b_2+a_2 \\ b_3 & a_3+c_3 & b_3+a_3 \end{vmatrix} + \begin{vmatrix} c_1 & a_1+c_1 & b_1+a_1 \\ c_2 & a_2+c_2 & b_2+a_2 \\ c_3 & a_3+c_3 & b_3+a_3 \end{vmatrix}$$

$$\stackrel{\text{①}}{=} \begin{vmatrix} c_1 & a_1 & b_1 \\ c_2 & a_2 & b_2 \\ c_3 & a_3 & b_3 \end{vmatrix} + \begin{vmatrix} b_1 & c_1 & a_1 \\ b_2 & c_2 & a_2 \\ b_3 & c_3 & a_3 \end{vmatrix} = \dots$$